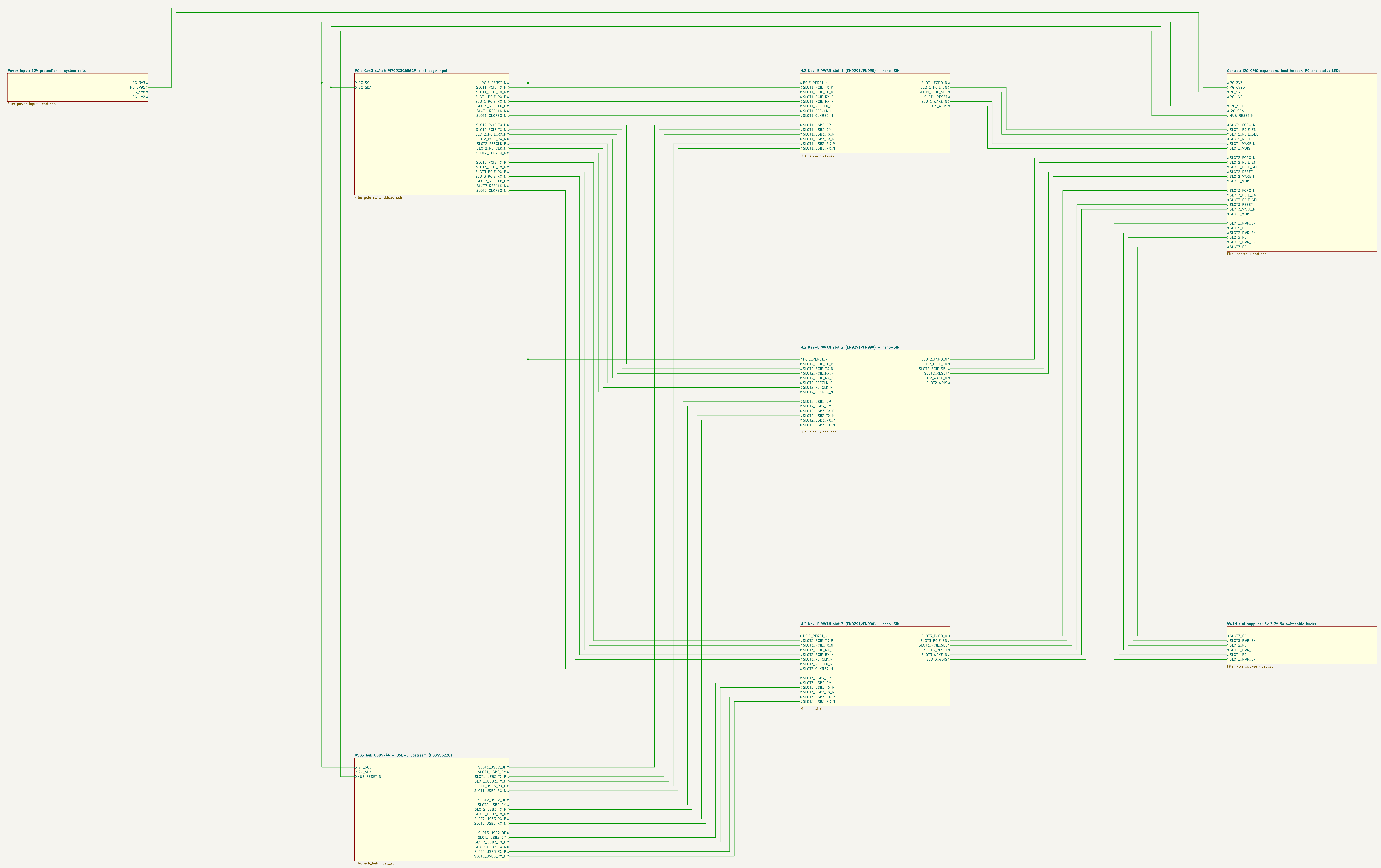
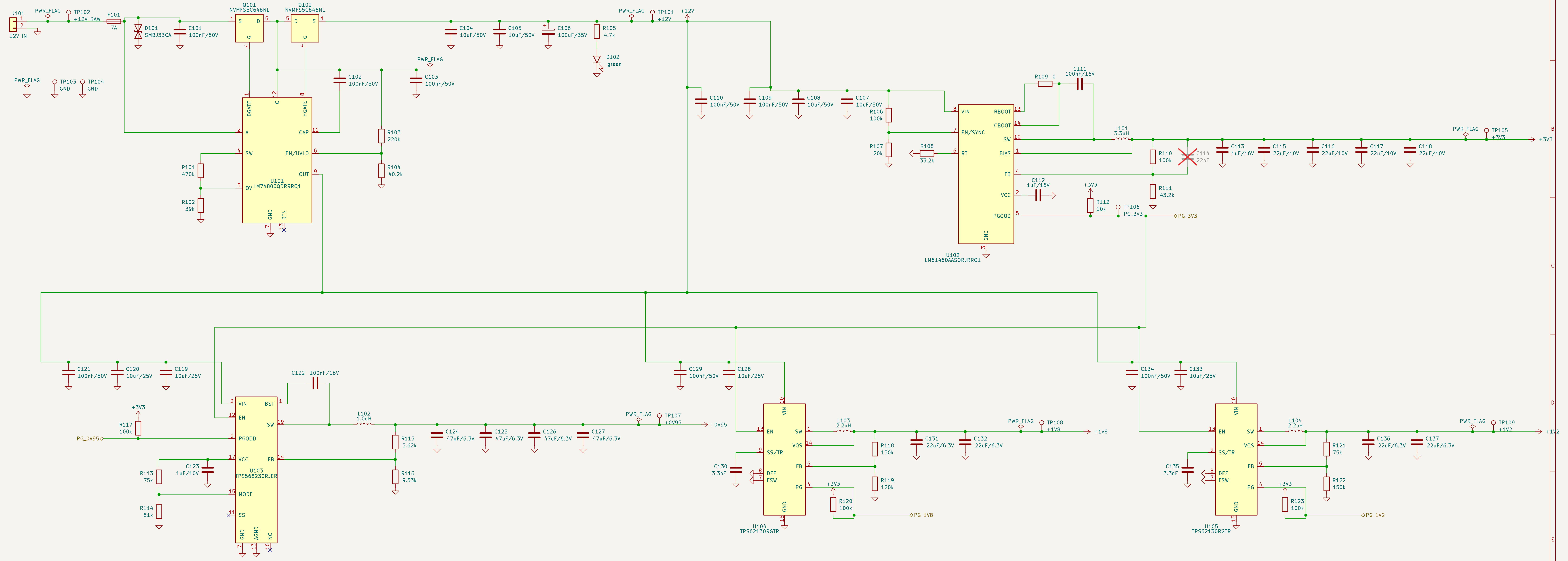

Power input: 12V protection + system rails
PG_3V3
PG_0V95
PG_1V8
PG_1V2
File: power_input.kicad_sch
PCIe Gen3 switch PI7C9X3G606GP + x1 edge input
I2C_SCL
I2C_SDA
PCIE_PERST_N
SLOT1_PCIE_TX_P
SLOT1_PCIE_TX_N
SLOT1_PCIE_RX_P
SLOT1_PCIE_RX_N
SLOT1_REFCLK_P
SLOT1_REFCLK_N
SLOT1_CLKREQ_N
SLOT2_PCIE_TX_P
SLOT2_PCIE_TX_N
SLOT2_PCIE_RX_P
SLOT2_PCIE_RX_N
SLOT2_REFCLK_P
SLOT2_REFCLK_N
SLOT2_CLKREQ_N
SLOT3_PCIE_TX_P
SLOT3_PCIE_TX_N
SLOT3_PCIE_RX_P
SLOT3_PCIE_RX_N
SLOT3_REFCLK_P
SLOT3_REFCLK_N
SLOT3_CLKREQ_N
File: pcie_switch.kicad_sch
M.2 Key–B WWAN slot 1 (EM9291/FN990) + nano–SIM
PCIE_PERST_N
SLOT1_PCIE_TX_P
SLOT1_PCIE_TX_N
SLOT1_PCIE_RX_P
SLOT1_PCIE_RX_N
SLOT1_REFCLK_P
SLOT1_REFCLK_N
SLOT1_CLKREQ_N
SLOT1_USB2_DP
SLOT1_USB2_DM
SLOT1_USB3_TX_P
SLOT1_USB3_TX_N
SLOT1_USB3_RX_P
SLOT1_USB3_RX_N
SLOT1_FCPO_N
SLOT1_PCIE_EN
SLOT1_PCIE_SEL
SLOT1_RESET
SLOT1_WAKE_N
SLOT1_WDIS
File: slot1.kicad_sch
Control: I2C GPIO expanders, host header, PG and status LEDs
PG_3V3
PG_0V95
PG_1V8
PG_1V2
I2C_SCL
I2C_SDA
HUB_RESET_N
SLOT1_FCPO_N
SLOT1_PCIE_EN
SLOT1_PCIE_SEL
SLOT1_RESET
SLOT1_WAKE_N
SLOT1_WDIS
SLOT2_FCPO_N
SLOT2_PCIE_EN
SLOT2_PCIE_SEL
SLOT2_RESET
SLOT2_WAKE_N
SLOT2_WDIS
SLOT3_FCPO_N
SLOT3_PCIE_EN
SLOT3_PCIE_SEL
SLOT3_RESET
SLOT3_WAKE_N
SLOT3_WDIS
SLOT1_PWR_EN
SLOT1_PG
SLOT2_PWR_EN
SLOT2_PG
SLOT3_PWR_EN
SLOT3_PG
File: control.kicad_sch
M.2 Key–B WWAN slot 2 (EM9291/FN990) + nano–SIM
PCIE_PERST_N
SLOT2_PCIE_TX_P
SLOT2_PCIE_TX_N
SLOT2_PCIE_RX_P
SLOT2_PCIE_RX_N
SLOT2_REFCLK_P
SLOT2_REFCLK_N
SLOT2_CLKREQ_N
SLOT2_USB2_DP
SLOT2_USB2_DM
SLOT2_USB3_TX_P
SLOT2_USB3_TX_N
SLOT2_USB3_RX_P
SLOT2_USB3_RX_N
SLOT2_FCPO_N
SLOT2_PCIE_EN
SLOT2_PCIE_SEL
SLOT2_RESET
SLOT2_WAKE_N
SLOT2_WDIS
File: slot2.kicad_sch
M.2 Key–B WWAN slot 3 (EM9291/FN990) + nano–SIM
PCIE_PERST_N
SLOT3_PCIE_TX_P
SLOT3_PCIE_TX_N
SLOT3_PCIE_RX_P
SLOT3_PCIE_RX_N
SLOT3_REFCLK_P
SLOT3_REFCLK_N
SLOT3_CLKREQ_N
SLOT3_USB2_DP
SLOT3_USB2_DM
SLOT3_USB3_TX_P
SLOT3_USB3_TX_N
SLOT3_USB3_RX_P
SLOT3_USB3_RX_N
SLOT3_FCPO_N
SLOT3_PCIE_EN
SLOT3_PCIE_SEL
SLOT3_RESET
SLOT3_WAKE_N
SLOT3_WDIS
File: slot3.kicad_sch
WWAN slot supplies: 3x 3.7V 6A switchable bucks
SLOT3_PG
SLOT3_PWR_EN
SLOT2_PG
SLOT2_PWR_EN
SLOT1_PG
SLOT1_PWR_EN
File: wwan_power.kicad_sch
USB3 hub USB5744 + USB–C upstream (HD3SS3220)
I2C_SCL
I2C_SDA
HUB_RESET_N
SLOT1_USB2_DP
SLOT1_USB2_DM
SLOT1_USB3_TX_P
SLOT1_USB3_TX_N
SLOT1_USB3_RX_P
SLOT1_USB3_RX_N
SLOT2_USB2_DP
SLOT2_USB2_DM
SLOT2_USB3_TX_P
SLOT2_USB3_TX_N
SLOT2_USB3_RX_P
SLOT2_USB3_RX_N
SLOT3_USB2_DP
SLOT3_USB2_DM
SLOT3_USB3_TX_P
SLOT3_USB3_TX_N
SLOT3_USB3_RX_P
SLOT3_USB3_RX_N
File: usb_hub.kicad_sch

Root sheet = wired system block diagram: every wire here is a hierarchical sheet pin connection (no global labels for signals).
Power rails (+12V, +3V3, +0V95, +1V8, +1V2, +3V7_S1..3) are distributed as power symbols inside the sheets.
PCIe: switch DS ports –> slots (Gen3 x1 + REFCLK + CLKREQ#), PERST# fan–out. USB: USB5744 DS ports 1–3 –> slots (USB2 + USB3).
Control: TCA6416A expanders –> slot FCPO#/RESET/W_DISABLE/PCIE_EN/PCIE_SEL(M.2 pin 20)/WAKE#, buck enables / PGOOD; host I2C to hub + PCIe switch.

Datasheet–verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto–routed); rails = power symbols; inter–sheet = hier. labels
Jorge Mendoza – WWAN carrier (spin 1 bench prototype)
Sheet: /
File: wwan–carrier.kicad_sch
Title: WWAN Carrier rev0.3: 3x M.2 WWAN, PCIe switch, USB3 hub
Size: A1
Date: 2026–10–10
Rev: 0.3
KiCad E.D.A. 9.0.2
Id: 1/9

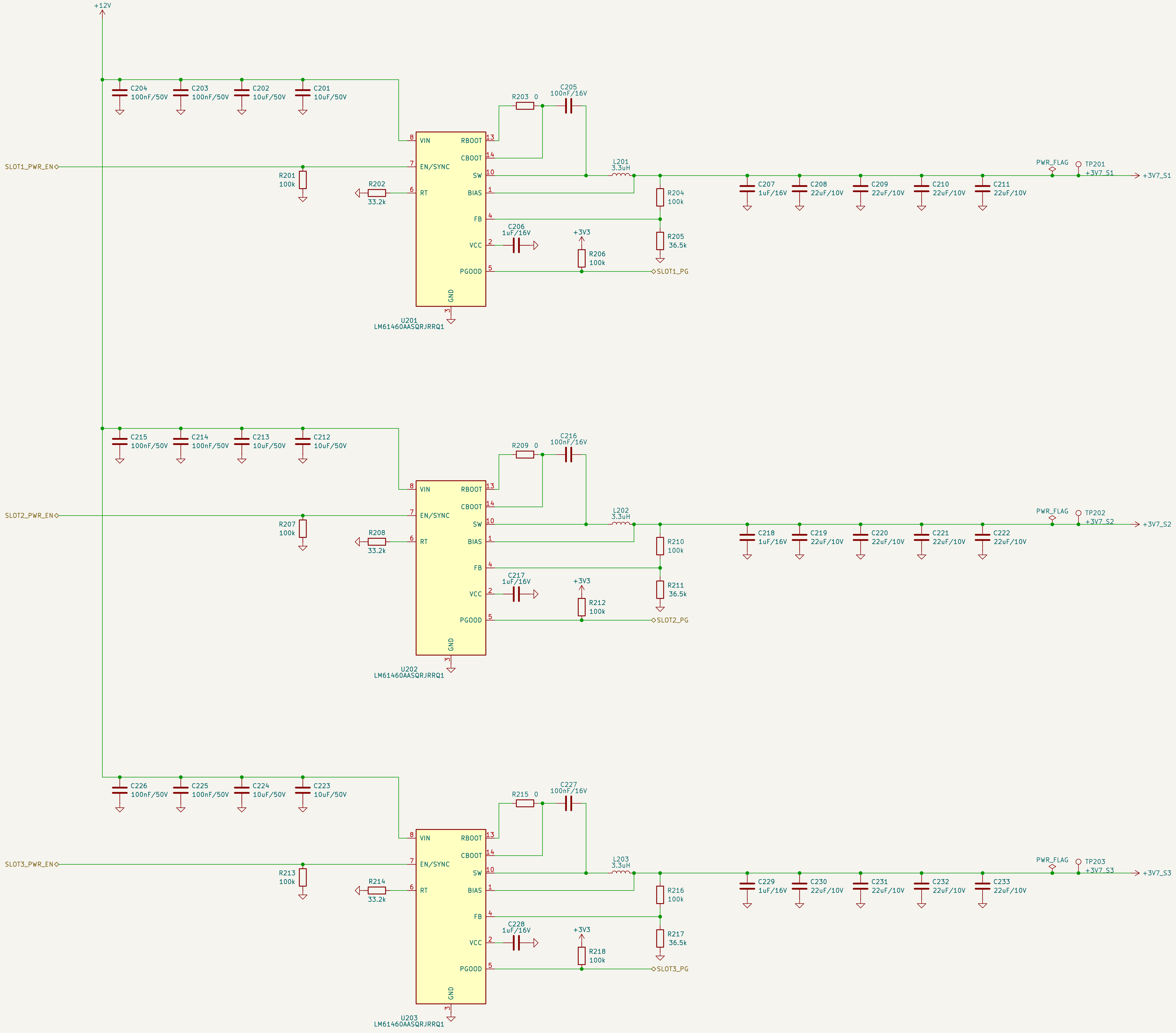

J101
12V IN
PWR_FLAG
TP102
+12V_RAW
F101
7A
D101
SMBJ33CA
C101
100nF/50V
Q101
NVMFS5C646NL
Q102
NVMFS5C646NL
PWR_FLAG
TP103
GND
TP104
GND
DGATE
C
HGATE
A
CAP
SW
EN/UVLO
OUT
OV
U101
LM74800QDRRRQ1
GND
RTN
R101
470k
R102
39k
C102
100nF/50V
C103
100nF/50V
R103
220k
R104
40.2k
C104
10uF/50V
C105
10uF/50V
C106
100uF/35V
R105
4.7k
D102
green
PWR_FLAG
TP101
+12V
+12V
C110
100nF/50V
C109
100nF/50V
C108
10uF/50V
C107
10uF/50V
R106
100k
R107
20k
R108
33.2k
VIN
EN/SYNC
RT
RBOOT
CBOOT
SW
BIAS
FB
VCC
PGOOD
GND
U102
LM61460AASQRJRRQ1
R109 0
C111
100nF/16V
L101
3.3uH
R110
100k
C114
22pF
R111
43.2k
C112
1uF/16V
+3V3
R112
10k
TP106
PG_3V3
PG_3V3
C113
1uF/16V
C115
22uF/10V
C116
22uF/10V
C117
22uF/10V
C118
22uF/10V
PWR_FLAG
TP105
+3V3
+3V3
C121
100nF/50V
C120
10uF/25V
C119
10uF/25V
C122 100nF/16V
VIN
EN
PGOOD
VCC
MODE
SS
BST
SW
FB
U103
TPS568230RJER
GND
AGND
NC
+3V3
R117
100k
PG_0V95
R113
75k
R114
51k
C123
1uF/10V
L102
1.0uH
R115
5.62k
R116
9.53k
C124
47uF/6.3V
C125
47uF/6.3V
C126
47uF/6.3V
C127
47uF/6.3V
PWR_FLAG
TP107
+0V95
+0V95
C129
100nF/50V
C128
10uF/25V
VIN
EN
SS/TR
DEF
FSW
SW
VOS
FB
PG
GND
U104
TPS62130RGTR
C130
3.3nF
L103
2.2uH
R118
150k
R119
120k
+3V3
R120
100k
PG_1V8
C131
22uF/6.3V
C132
22uF/6.3V
PWR_FLAG
TP108
+1V8
+1V8
C134
100nF/50V
C133
10uF/25V
U105
TPS62130RGTR
C135
3.3nF
L104
2.2uH
R121
75k
R122
150k
+3V3
R123
100k
PG_1V2
C136
22uF/6.3V
C137
22uF/6.3V
PWR_FLAG
TP109
+1V2
+1V2

12V INPUT PROTECTION (LM74800–Q1, common–drain back–to–back NFETs, per TI LM7480–Q1 DS SNOSD95C Fig.10–1)
– Q101 (DGATE) = ideal diode / reverse–battery; Q102 (HGATE) = overvoltage cut–off.
– OV cut–off = 1.231V x (470k+39k)/39k = ~16.1V (protects TPS568230 18V / TPS62130 17V abs–max downstream).
– UVLO rising = 1.231V x (220k+40.2k)/40.2k = ~8.0V.
– LM7480 exposed pad (RTN) must be left FLOATING – do not tie to GND copper (DS Table 6–1).
– Fuse/TVS/FET MPNs are proposals – VERIFY ratings and inrush (3x(2x220uF)+bulk) before ordering.
SYSTEM RAILS
– +3V3: LM61460–Q1 (substitute for backordered LMQ61460–Q1; same RJR–14 pinout per both TI datasheets), Vout=1V x (1+100k/43.2k)=3.31V, 400kHz.
– +0V95 (PI7C9X3G606GP VDDC/VP/C_VDDC, worst ~3.3A @Tj80C per DS Table 12–12): TPS568230, Vref 0.6V, 5.62k/9.53k –> 0.954V, FCCM 800kHz.
– +1V8 (PI7C VDDR/VPH/C_VDDR ~0.25A + EEPROM + PCA9306): TPS62130, 150k/120k –> 1.80V.
– +1V2 (USB5744 VDD12, 688mA typ with 4 SS ports per DS Table 10–3): TPS62130, 75k/150k –> 1.20V.
– Low–voltage regulators enabled by PG_3V3 (simple sequencing; PI7C requires PERST# held until rails+REFCLK stable).

Datasheet–verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto–routed); rails = power symbols; inter–sheet = hier. labels
Jorge Mendoza – WWAN carrier (spin 1 bench prototype)
Sheet: /Power input: 12V protection + system rails/
File: power_input.kicad_sch

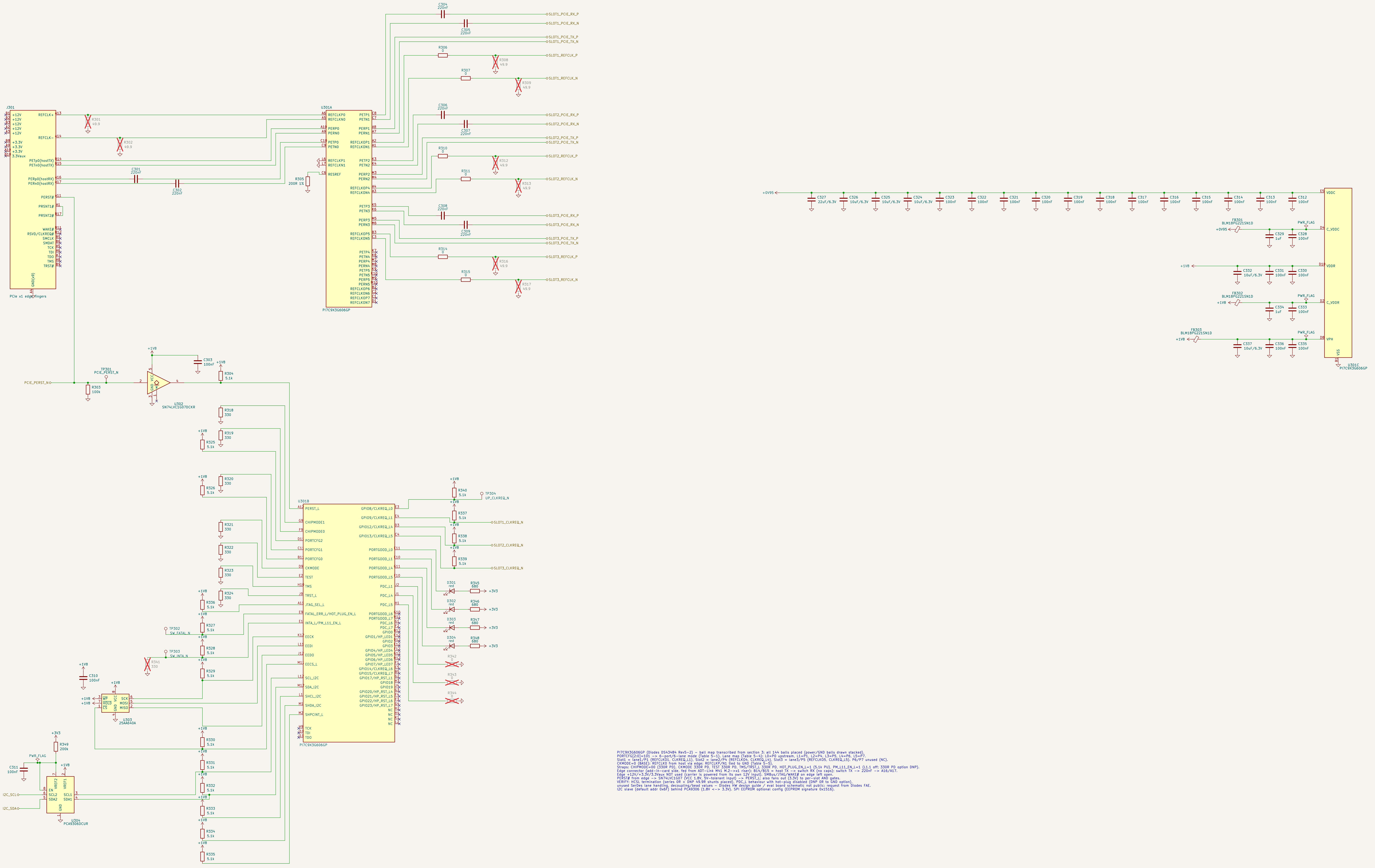
Title: Power input: 12V protection + system rails

Size: A2
Date: 2026–10–10
Rev: 0.3
KiCad E.D.A. 9.0.2
Id: 2/9


+12V
C204
100nF/50V
C203
100nF/50V
C202
10uF/50V
C201
10uF/50V
R203 0
C205
100nF/16V
SLOT1_PWR_EN
R201
100k
R202
33.2k
VIN
EN/SYNC
RT
RBOOT
CBOOT
SW
BIAS
FB
VCC
PGOOD
GND
L201
3.3uH
R204
100k
R205
36.5k
C206
1uF/16V
+3V3
R206
100k
SLOT1_PG
C207
1uF/16V
C208
22uF/10V
C209
22uF/10V
C210
22uF/10V
C211
22uF/10V
PWR_FLAG
TP201
+3V7_S1
+3V7_S1
U201
LM61460AASQRJRRQ1
C215
100nF/50V
C214
100nF/50V
C213
10uF/50V
C212
10uF/50V
R209 0
C216
100nF/16V
SLOT2_PWR_EN
R207
100k
R208
33.2k
L202
3.3uH
R210
100k
R211
36.5k
C217
1uF/16V
+3V3
R212
100k
SLOT2_PG
C218
1uF/16V
C219
22uF/10V
C220
22uF/10V
C221
22uF/10V
C222
22uF/10V
PWR_FLAG
TP202
+3V7_S2
+3V7_S2
U202
LM61460AASQRJRRQ1
C226
100nF/50V
C225
100nF/50V
C224
10uF/50V
C223
10uF/50V
R215 0
C227
100nF/16V
SLOT3_PWR_EN
R213
100k
R214
33.2k
L203
3.3uH
R216
100k
R217
36.5k
C228
1uF/16V
+3V3
R218
100k
SLOT3_PG
C229
1uF/16V
C230
22uF/10V
C231
22uF/10V
C232
22uF/10V
C233
22uF/10V
PWR_FLAG
TP203
+3V7_S3
+3V7_S3
U203
LM61460AASQRJRRQ1

3x LM61460-Q1 (6A) – one per M.2 slot. Module VCC window: Sierra EM92XX 3.135–4.4V, peak 3.0A, ripple <100mVpp 1–100kHz (PTS 4.3.3);
Telit FN990 VPH_PWR 3.135–4.4V, peak 4A, ripple max 30mV (HDG Rev 3 Tab.15). Both: VCC must stay >=3.3V for EM9293 GNSS bias.
Vout = 1.0V x (1 + 100k/36.5k) = 3.74V (headroom for connector/trace drop). fsw 400kHz (RT 33.2k), L = 3.3uH XAL7070.
EN driven by TCA6416A (control sheet); 100k pull–down keeps slot OFF at power–up. PGOOD –> expander + PG LED.
Slot bulk (2x220uF low–ESR + 1uF/100nF/pF HF caps; FN990 HDG recommends >=100uF low–ESR at the card) is on each slot sheet.
NOTE: no active output discharge – after EN low allow bleed–down before re–enable (power–cycle = sysfs remove –> EN off –> wait –> EN on –> rescan).

Datasheet–verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto–routed); rails = power symbols; inter–sheet = hier. labels
Jorge Mendoza – WWAN carrier (spin 1 bench prototype)
Sheet: /WWAN slot supplies: 3x 3.7V 6A switchable bucks/
File: wwan_power.kicad_sch
Title: WWAN slot supplies: 3x 3.7V 6A switchable bucks
Size: A2 | Date: 2026–10–10 | **Rev: 0.3**
KiCad E.D.A. 9.0.2 | Id: 3/9

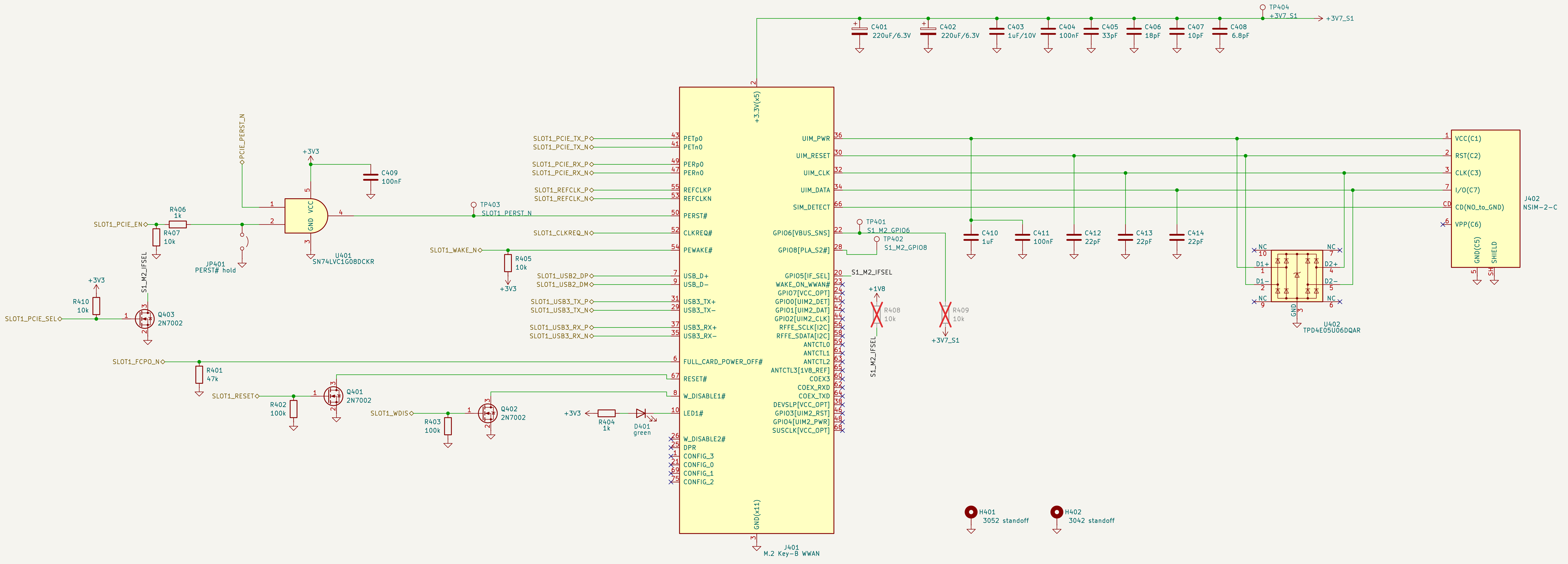

J301
+12V
+3.3V
3.3Vaux
REFCLK+
REFCLK−
PETp0(hostTX)
PETn0(hostTX)
PERp0(hostRX)
PERn0(hostRX)
PERST#
PRSNT1#
PRSNT2#
WAKE#
RSVD/CLKREQ#
SMCLK
SMDAT
TCK
TDI
TDO
TMS
TRST#
GND(x9)
PCIe x1 edge fingers
R301 49.9
R302 49.9
C301 220nF
C302 220nF
U301A
REFCLKP0
REFCLKN0
PERP0
PERN0
PETP0
PETN0
REFCLKP1
REFCLKN1
RESREF
R305 200R 1%
PETP1
PETN1
PERP1
PERN1
REFCLKOP1
REFCLKON1
PETP2
PETN2
PERP2
PERN2
REFCLKOP4
REFCLKON4
PETP3
PETN3
PERP3
PERN3
REFCLKOP5
REFCLKON5
PETP4
PETN4
PERP4
PERN4
PETP5
PETN5
PERP5
PERN5
REFCLKOP6
REFCLKON6
REFCLKOP7
REFCLKON7
PI7C9X3G606GP
C304 220nF
C305 220nF
R306 0
R307 0
R308 49.9
R309 49.9
SLOT1_PCIE_RX_P
SLOT1_PCIE_RX_N
SLOT1_PCIE_TX_P
SLOT1_PCIE_TX_N
SLOT1_REFCLK_P
SLOT1_REFCLK_N
C306 220nF
C307 220nF
R310 0
R311 0
R312 49.9
R313 49.9
SLOT2_PCIE_RX_P
SLOT2_PCIE_RX_N
SLOT2_PCIE_TX_P
SLOT2_PCIE_TX_N
SLOT2_REFCLK_P
SLOT2_REFCLK_N
C308 220nF
C309 220nF
R314 0
R315 0
R316 49.9
R317 49.9
SLOT3_PCIE_RX_P
SLOT3_PCIE_RX_N
SLOT3_PCIE_TX_P
SLOT3_PCIE_TX_N
SLOT3_REFCLK_P
SLOT3_REFCLK_N
+0V95
C327 22uF/6.3V
C326 10uF/6.3V
C325 10uF/6.3V
C324 10uF/6.3V
C323 100nF
C322 100nF
C321 100nF
C320 100nF
C319 100nF
C318 100nF
C317 100nF
C316 100nF
C315 100nF
C314 100nF
C313 100nF
C312 100nF
VDDC
FB301 BLM18PG221SN1D
C329 1uF
C328 100nF
PWR_FLAG
C_VDDC
+1V8
C332 10uF/6.3V
C331 100nF
C330 100nF
VDDR
FB302 BLM18PG221SN1D
C334 1uF
C333 100nF
C_VDDR
FB303 BLM18PG221SN1D
C337 10uF/6.3V
C336 100nF
C335 100nF
VPH
VSS
U301C
PI7C9X3G606GP
PCIE_PERST_N
TP301 PCIE_PERST_N
R303 100k
U302 SN74LVC1G07DCKR
C303 100nF
R304 5.1k
R318 330
R319 330
R325 5.1k
R320 330
R326 5.1k
R321 330
R322 330
R323 330
R324 330
R336 5.1k
R327 5.1k
TP302 SW_FATAL_N
R328 5.1k
TP303 SW_INTA_N
R341 330
R329 5.1k
C310 100nF
U303 25AA640A
WP
HOLD
CS
SCK
MOSI
MISO
R330 5.1k
R331 5.1k
R332 5.1k
R333 5.1k
R334 5.1k
R335 5.1k
+3V3
R349 200k
C311 100nF
U304 PCA9306DCUR
EN
VREF2
VREF1
SCL2
SDA2
SCL1
SDA1
I2C_SCL
I2C_SDA
U301B
PERST_L
CHIPMODE1
CHIPMODE0
PORTCFG2
PORTCFG1
PORTCFG0
CKMODE
TEST
TMS
TRST_L
JTAG_SEL_L
FATAL_ERR_L/HOT_PLUG_EN_L
INTA_L/PM_L11_EN_L
EECK
EEDI
EEDO
EECS_L
SCL_I2C
SDA_I2C
SHCL_I2C
SHDA_I2C
SHPCINT_L
TCK
TDI
TDO
GPIO8/CLKREQ_L0
GPIO9/CLKREQ_L1
GPIO12/CLKREQ_L4
GPIO13/CLKREQ_L5
PORTGOOD_L0
PORTGOOD_L1
PORTGOOD_L4
PORTGOOD_L5
PDC_L1
PDC_L4
PDC_L5
PORTGOOD_L6
PORTGOOD_L7
PDC_L6
PDC_L7
GPIO0
GPIO1/HP_LED1
GPIO2
GPIO3
GPIO4/HP_LED4
GPIO5/HP_LED5
GPIO6/HP_LED6
GPIO7/HP_LED7
GPIO14/CLKREQ_L6
GPIO15/CLKREQ_L7
GPIO17/HP_RST_L1
GPIO18
GPIO19
GPIO20/HP_RST_L4
GPIO21/HP_RST_L5
GPIO22/HP_RST_L6
GPIO23/HP_RST_L7
NC
PI7C9X3G606GP
R340 5.1k
TP304 UP_CLKREQ_N
R337 5.1k
SLOT1_CLKREQ_N
R338 5.1k
SLOT2_CLKREQ_N
R339 5.1k
SLOT3_CLKREQ_N
D301 red
D302 red
D303 red
D304 red
R345 680
R346 680
R347 680
R348 680
R342 0
R343 0
R344 0

PI7C9X3G606GP (Diodes DS43484 Rev5−2) − ball map transcribed from section 3; all 144 balls placed (power/GND balls drawn stacked).
PORTCFG[2:0]=101 −> 6−port/6−lane mode (Table 5−1). Lane map (Table 5−4): L0=P0 upstream, L1=P1, L2=P4, L3=P5, L4=P6, L5=P7.
Slot1 = lane1/P1 (REFCLKO1, CLKREQ_L1), Slot2 = lane2/P4 (REFCLKO4, CLKREQ_L4), Slot3 = lane3/P5 (REFCLKO5, CLKREQ_L5). P6/P7 unused (NC).
CKMODE=0 (BASE): REFCLK0 from host via edge; REFCLKP/N1 tied to GND (Table 5−5).
Straps: CHIPMODE=00 (330R PD), CKMODE 330R PD, TEST 330R PD, TMS/TRST_L 330R PD, HOT_PLUG_EN_L=1 (5.1k PU), PM_L11_EN_L=1 (L1.1 off; 330R PD option DNP).
Edge connector (add−in−card side, fed from ADT−Link M41 M.2−>x1 riser): B14/B15 = host TX −> switch RX (no caps); switch TX −> 220nF −> A16/A17.
Edge +12V/+3.3V/3.3Vaux NOT used (carrier is powered from its own 12V input). SMBus/JTAG/WAKE# on edge left open.
PERST# from edge −> SN74LVC1G07 (VCC 1.8V, 5V−tolerant input) −> PERST_L; also fans out (3.3V) to per−slot AND gates.
VERIFY: HCSL termination (series 0R + DNP 49.9R shunts placed), PDC_L behaviour with hot−plug disabled (DNP 0R to GND option),
unused SerDes lane handling, decoupling/bead values − Diodes HW design guide / eval board schematic not public: request from Diodes FAE.
I2C slave (default addr 0x6F) behind PCA9306 (1.8V <−> 3.3V). SPI EEPROM optional config (EEPROM signature 0x1516).

Datasheet−verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto−routed); rails = power symbols; inter−sheet = hier. labels
Jorge Mendoza − WWAN carrier (spin 1 bench prototype)
Sheet: /PCIe Gen3 switch PI7C9X3G606GP + x1 edge input/
File: pcie_switch.kicad_sch

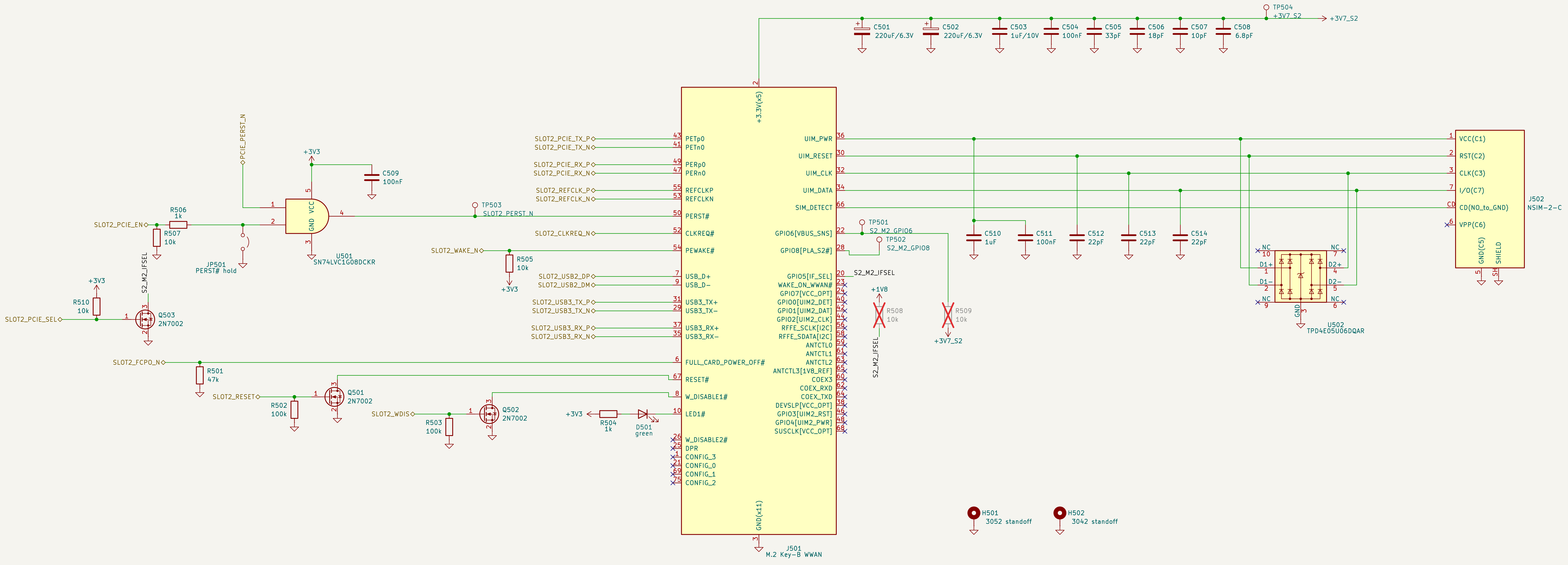
Title: PCIe Gen3 switch PI7C9X3G606GP + x1 edge input

Size: A1
Date: 2026−10−10
Rev: 0.3
KiCad E.D.A. 9.0.2
Id: 4/9


C401
220uF/6.3V
C402
220uF/6.3V
C403
1uF/10V
C404
100nF
C405
33pF
C406
18pF
C407
10pF
C408
6.8pF
TP404
+3V7_S1
+3V7_S1
PCIE_PERST_N
+3V3
C409
100nF
R406
1k
SLOT1_PCIE_EN
R407
10k
JP401
PERST# hold
U401
SN74LVC1G08DCKR
VCC
GND
TP403
SLOT1_PERST_N
S1_M2_IFSEL
+3V3
R410
10k
SLOT1_PCIE_SEL
Q403
2N7002
SLOT1_FCPO_N
R401
47k
SLOT1_RESET
R402
100k
Q401
2N7002
SLOT1_WDIS
R403
100k
Q402
2N7002
+3V3
R404
1k
D401
green
SLOT1_WAKE_N
R405
10k
+3V3
SLOT1_PCIE_TX_P
SLOT1_PCIE_TX_N
SLOT1_PCIE_RX_P
SLOT1_PCIE_RX_N
SLOT1_REFCLK_P
SLOT1_REFCLK_N
SLOT1_CLKREQ_N
SLOT1_USB2_DP
SLOT1_USB2_DM
SLOT1_USB3_TX_P
SLOT1_USB3_TX_N
SLOT1_USB3_RX_P
SLOT1_USB3_RX_N
+3.3V(x5)
PETp0
PETn0
PERp0
PERn0
REFCLKP
REFCLKN
PERST#
CLKREQ#
PEWAKE#
USB_D+
USB_D−
USB3_TX+
USB3_TX−
USB3_RX+
USB3_RX−
FULL_CARD_POWER_OFF#
RESET#
W_DISABLE1#
LED1#
W_DISABLE2#
DPR
CONFIG_3
CONFIG_0
CONFIG_1
CONFIG_2
UIM_PWR
UIM_RESET
UIM_CLK
UIM_DATA
SIM_DETECT
GPIO6[VBUS_SNS]
GPIO8[PLA_S2#]
GPIO5[IF_SEL]
WAKE_ON_WWAN#
GPIO7[VCC_OPT]
GPIO0[UIM2_DET]
GPIO1[UIM2_DAT]
GPIO2[UIM2_CLK]
RFFE_SCLK[I2C]
RFFE_SDATA[I2C]
ANTCTL0
ANTCTL1
ANTCTL2
ANTCTL3[1V8_REF]
COEX3
COEX_RXD
COEX_TXD
DEVSLP[VCC_OPT]
GPIO3[UIM2_RST]
GPIO4[UIM2_PWR]
SUSCLK[VCC_OPT]
GND(x11)
J401
M.2 Key−B WWAN
TP401
S1_M2_GPIO6
TP402
S1_M2_GPIO8
S1_M2_IFSEL
+1V8
R408
10k
S1_M2_IFSEL
R409
10k
+3V7_S1
C410
1uF
C411
100nF
C412
22pF
C413
22pF
C414
22pF
NC
D1+
D1−
D2+
D2−
GND
U402
TPD4E05U06DQAR
VCC(C1)
RST(C2)
CLK(C3)
I/O(C7)
CD(NO_to_GND)
VPP(C6)
GND(C5)
SHIELD
J402
NSIM−2−C
H401
3052 standoff
H402
3042 standoff

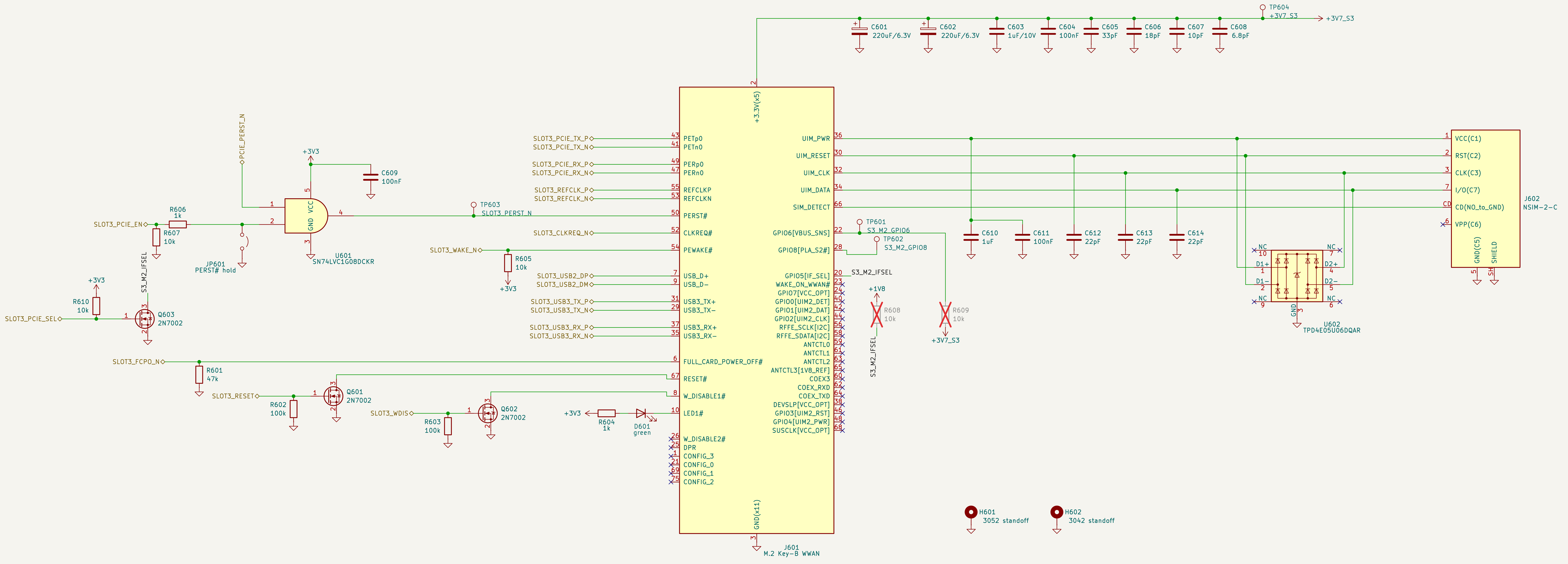
SLOT 1: generic M.2 3052/3042 Key−B WWAN socket. Pins checked vs Sierra EM92XX PTS Rev 5 (Jan 2025) + Telit FN990 HDG Rev 3.
PCIe Gen3 x1 to switch port 1 (lane1); USB2 + USB3 to USB5744 port 1. Module TX AC caps on module.
FULL_CARD_POWER_OFF# (pin 6): TCA6416A 3.3V push−pull + 47k PD (EM92 VIH 1.35−4.4V; FN990 1.8V/VPH domain, 47k internal PD) −> module off by default.
  Firmware: drive FCPO# LOW before removing +3V7 (no high levels into an unpowered card − FN990 HDG 6.2.1 back−power note).
RESET# (pin 67) and W_DISABLE1# (pin 8): open−drain 2N7002 only (both vendors forbid pull−up/push−pull on RESET#). Pulse >=1s covers EM92 (450ms) + FN990 (1s).
PERST# = PCIE_PERST_N AND S_PCIE_EN_L (SN74LVC1G08, 3.3V; EM92 VIH 1.5V..VCC, FN990 VPH domain). 10k PD (rev 0.3, was PU) keeps PERST# low until
  firmware sets PCIE_EN >=150ms after the slot rail is valid (EM92 T_pvpgl; no host signal into the card for the first 100ms). JP = hard PERST# hold.
Pin 20 host−IF select: NFET pulls pin 20 LOW while SLOTn_PCIE_SEL is HIGH (10k PU = default PCIe). Own expander line (U802 ex−spare), NOT PCIE_EN:
  pin 20 is sampled at boot, while PERST# must stay gated low until >=150ms after power valid (EM92 T_pvpgl) − one line cannot do both.
  FN990 USB_PCIE_SWITCH (int. 10k PU to 1.8V): LOW = PCIe (+USB debug), HIGH = USB only. EM92 PCIE_DIS (PD, max 2.1V): LOW/open = PCIe, HIGH = USB
  −> fit the DNP 10k to +1V8 only to let PCIE_SEL=0 select USB on a Sierra card. Changing mode needs a module power cycle.
Pin 22: EM92 VBUS_SENSE (PD, 1.6−5.25V = USB present) / FN990 TGPIO_06 (1.8V). DNP 10k to slot rail: FIT ONLY for Sierra; NEVER with Telit.
Pin 28: EM92 PLA_S2_N (power−loss ack, LOW = safe to remove power) / FN990 TGPIO_08 −> test point (1.8V; not level−shifted in rev 0.3).
Pins 24/38/68 NC (EM92 optional VCC; FN990 TGPIO). Pin 73 = GND (both vendors) −> 3.3V sideband. 40−48 (UIM2), 56/58, 59−65, 23, 25, 26, CONFIG_x: NC.
SIM: CUI NSIM−2−C card−detect closes to GND when card present. FN990 UIM1_PRESENT: active−low OK. EM92 SIM1_DETECT default 0V=absent: set polarity by AT
  (configurable per PTS Table 2−1 note e) or disable hot−swap detect. ESD + 22pF filters near holder.
GNSS / NTN antennas: NO carrier RF or GNSS bias circuit. All antennas are off−board pigtails on the card U.FL/MHF4 ports (see docs/antenna−gnss−plan.md).
  FN990: internal GNSS bias LDO (2.9−3.15V, 100mA) on its GNSS port − active antenna OK. EM9291: GNSS on shared ANT1 − DC on ANT0−3 damages it: PASSIVE only.
Mechanical: standoff holes for 3052 (52mm) and 3042 (42mm) cards.

Datasheet−verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto−routed); rails = power symbols; inter−sheet = hier. labels
Jorge Mendoza − WWAN carrier (spin 1 bench prototype)
Sheet: /M.2 Key−B WWAN slot 1 (EM9291/FN990) + nano−SIM/
File: slot1.kicad_sch
Title: M.2 Key−B WWAN slot 1 (EM9291/FN990) + nano−SIM
Size: A2
Date: 2026−10−10
Rev: 0.3
KiCad E.D.A. 9.0.2
Id: 5/9


TP504
+3V7_S2
+3V7_S2
C501
220uF/6.3V
C502
220uF/6.3V
C503
1uF/10V
C504
100nF
C505
33pF
C506
18pF
C507
10pF
C508
6.8pF
PCIE_PERST_N
+3V3
C509
100nF
R506
1k
SLOT2_PCIE_EN
R507
10k
U501
SN74LVC1G08DCKR
GND VCC
JP501
PERST# hold
TP503
SLOT2_PERST_N
S2_M2_IFSEL
+3V3
R510
10k
SLOT2_PCIE_SEL
Q503
2N7002
SLOT2_PCIE_TX_P
SLOT2_PCIE_TX_N
SLOT2_PCIE_RX_P
SLOT2_PCIE_RX_N
SLOT2_REFCLK_P
SLOT2_REFCLK_N
SLOT2_CLKREQ_N
SLOT2_WAKE_N
R505
10k
+3V3
SLOT2_USB2_DP
SLOT2_USB2_DM
SLOT2_USB3_TX_P
SLOT2_USB3_TX_N
SLOT2_USB3_RX_P
SLOT2_USB3_RX_N
SLOT2_FCPO_N
R501
47k
SLOT2_RESET
R502
100k
Q501
2N7002
SLOT2_WDIS
R503
100k
Q502
2N7002
+3V3
R504
1k
D501
green
+3.3V(x5)
PETp0
PETn0
PERp0
PERn0
REFCLKP
REFCLKN
PERST#
CLKREQ#
PEWAKE#
USB_D+
USB_D−
USB3_TX+
USB3_TX−
USB3_RX+
USB3_RX−
FULL_CARD_POWER_OFF#
RESET#
W_DISABLE1#
LED1#
W_DISABLE2#
DPR
CONFIG_3
CONFIG_0
CONFIG_1
CONFIG_2
UIM_PWR
UIM_RESET
UIM_CLK
UIM_DATA
SIM_DETECT
GPIO6[VBUS_SNS]
GPIO8[PLA_S2#]
GPIO5[IF_SEL]
WAKE_ON_WWAN#
GPIO7[VCC_OPT]
GPIO0[UIM2_DET]
GPIO1[UIM2_DAT]
GPIO2[UIM2_CLK]
RFFE_SCLK[I2C]
RFFE_SDATA[I2C]
ANTCTL0
ANTCTL1
ANTCTL2
ANTCTL3[1V8_REF]
COEX3
COEX_RXD
COEX_TXD
DEVSLP[VCC_OPT]
GPIO3[UIM2_RST]
GPIO4[UIM2_PWR]
SUSCLK[VCC_OPT]
GND(x11)
J501
M.2 Key−B WWAN
TP501
S2_M2_GPIO6
TP502
S2_M2_GPIO8
S2_M2_IFSEL
+1V8
R508
10k
S2_M2_IFSEL
R509
10k
+3V7_S2
C510
1uF
C511
100nF
C512
22pF
C513
22pF
C514
22pF
NC
D1+
D1−
NC
NC
D2+
D2−
NC
GND
U502
TPD4E05U06DQAR
VCC(C1)
RST(C2)
CLK(C3)
I/O(C7)
CD(NO_to_GND)
VPP(C6)
GND(C5)
SHIELD
J502
NSIM−2−C
H501
3052 standoff
H502
3042 standoff

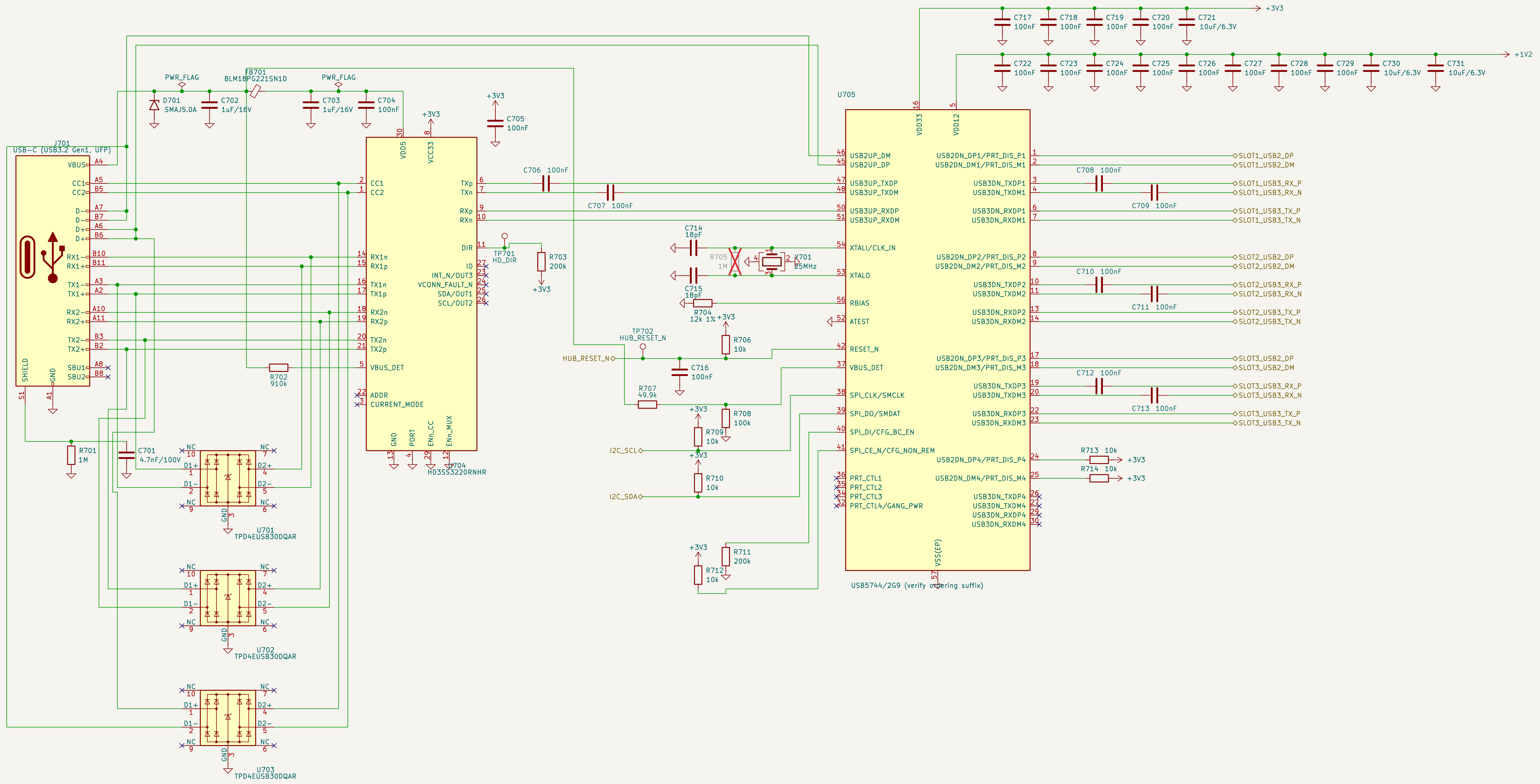
SLOT 2: generic M.2 3052/3042 Key−B WWAN socket. Pins checked vs Sierra EM92XX PTS Rev 5 (Jan 2025) + Telit FN990 HDG Rev 3.
PCIe Gen3 x1 to switch port 4 (lane2); USB2 + USB3 to USB5744 port 2. Module TX AC caps on module.
FULL_CARD_POWER_OFF# (pin 6): TCA6416A 3.3V push−pull + 47k PD (EM92 VIH 1.35−4.4V; FN990 1.8V/VPH domain, 47k internal PD) −> module off by default.
  Firmware: drive FCPO# LOW before removing +3V7 (no high levels into an unpowered card − FN990 HDG 6.2.1 back−power note).
RESET# (pin 67) and W_DISABLE1# (pin 8): open−drain 2N7002 only (both vendors forbid pull−up/push−pull on RESET#). Pulse >=1s covers EM92 (450ms) + FN990 (1s).
PERST# = PCIE_PERST_N AND S_PCIE_EN_L (SN74LVC1G08, 3.3V; EM92 VIH 1.5V..VCC, FN990 VPH domain). 10k PD (rev 0.3, was PU) keeps PERST# low until
  firmware sets PCIE_EN >=150ms after the slot rail is valid (EM92 T_pvpgl; no host signal into the card for the first 100ms). JP = hard PERST# hold.
Pin 20 host−IF select: NFET pulls pin 20 LOW while SLOTn_PCIE_SEL is HIGH (10k PU = default PCIe). Own expander line (U802 ex−spare), NOT PCIE_EN:
  pin 20 is sampled at boot, while PERST# must stay gated low until >=150ms after power valid (EM92 T_pvpgl) − one line cannot do both.
  FN990 USB_PCIE_SWITCH (int. 10k PU to 1.8V): LOW = PCIe (+USB debug), HIGH = USB only. EM92 PCIE_DIS (PD, max 2.1V): LOW/open = PCIe, HIGH = USB
  −> fit the DNP 10k to +1V8 only to let PCIE_SEL=0 select USB on a Sierra card. Changing mode needs a module power cycle.
Pin 22: EM92 VBUS_SENSE (PD, 1.6−5.25V = USB present) / FN990 TGPIO_06 (1.8V). DNP 10k to slot rail: FIT ONLY for Sierra; NEVER with Telit.
Pin 28: EM92 PLA_S2_N (power−loss ack, LOW = safe to remove power) / FN990 TGPIO_08 −> test point (1.8V; not level−shifted in rev 0.3).
Pins 24/38/68 NC (EM92 optional VCC; FN990 TGPIO). Pin 73 = GND (both vendors) −> 3.3V sideband. 40−48 (UIM2), 56/58, 59−65, 23, 25, 26, CONFIG_x: NC.
SIM: CUI NSIM−2−C card−detect closes to GND when card present. FN990 UIM1_PRESENT: active−low OK. EM92 SIM1_DETECT default 0V=absent: set polarity by AT
  (configurable per PTS Table 2−1 note e) or disable hot−swap detect. ESD + 22pF filters near holder.
GNSS / NTN antennas: NO carrier RF or GNSS bias circuit. All antennas are off−board pigtails on the card U.FL/MHF4 ports (see docs/antenna−gnss−plan.md).
  FN990: internal GNSS bias LDO (2.9−3.15V, 100mA) on its GNSS port − active antenna OK. EM9291: GNSS on shared ANT1 − DC on ANT0−3 damages it: PASSIVE only.
Mechanical: standoff holes for 3052 (52mm) and 3042 (42mm) cards.

Datasheet−verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto−routed); rails = power symbols; inter−sheet = hier. labels
Jorge Mendoza − WWAN carrier (spin 1 bench prototype)
Sheet: /M.2 Key−B WWAN slot 2 (EM9291/FN990) + nano−SIM/
File: slot2.kicad_sch
Title: M.2 Key−B WWAN slot 2 (EM9291/FN990) + nano−SIM
Size: A2 | Date: 2026−10−10 | **Rev: 0.3**
KiCad E.D.A. 9.0.2
Id: 6/9


C601 220uF/6.3V
C602 220uF/6.3V
C603 1uF/10V
C604 100nF
C605 33pF
C606 18pF
C607 10pF
C608 6.8pF
TP604 +3V7_S3
+3V7_S3
PCIE_PERST_N
+3V3
C609 100nF
SLOT3_PCIE_EN
R606 1k
R607 10k
JP601 PERST# hold
U601 SN74LVC1G08DCKR
GND VCC
TP603 SLOT3_PERST_N
S3_M2_IFSEL
+3V3
R610 10k
SLOT3_PCIE_SEL
Q603 2N7002
SLOT3_WAKE_N
R605 10k
+3V3
SLOT3_FCPO_N
R601 47k
SLOT3_RESET
R602 100k
Q601 2N7002
SLOT3_WDIS
R603 100k
Q602 2N7002
+3V3
R604 1k
D601 green
SLOT3_PCIE_TX_P
SLOT3_PCIE_TX_N
SLOT3_PCIE_RX_P
SLOT3_PCIE_RX_N
SLOT3_REFCLK_P
SLOT3_REFCLK_N
SLOT3_CLKREQ_N
SLOT3_USB2_DP
SLOT3_USB2_DM
SLOT3_USB3_TX_P
SLOT3_USB3_TX_N
SLOT3_USB3_RX_P
SLOT3_USB3_RX_N
+3.3V(x5)
PETp0
PETn0
PERp0
PERn0
REFCLKP
REFCLKN
PERST#
CLKREQ#
PEWAKE#
USB_D+
USB_D−
USB3_TX+
USB3_TX−
USB3_RX+
USB3_RX−
FULL_CARD_POWER_OFF#
RESET#
W_DISABLE1#
LED1#
W_DISABLE2#
DPR
CONFIG_3
CONFIG_0
CONFIG_1
CONFIG_2
UIM_PWR
UIM_RESET
UIM_CLK
UIM_DATA
SIM_DETECT
GPIO6[VBUS_SNS]
GPIO8[PLA_S2#]
GPIO5[IF_SEL]
WAKE_ON_WWAN#
GPIO7[VCC_OPT]
GPIO0[UIM2_DET]
GPIO1[UIM2_DAT]
GPIO2[UIM2_CLK]
RFFE_SCLK[I2C]
RFFE_SDATA[I2C]
ANTCTL0
ANTCTL1
ANTCTL2
ANTCTL3[1V8_REF]
COEX3
COEX_RXD
COEX_TXD
DEVSLP[VCC_OPT]
GPIO3[UIM2_RST]
GPIO4[UIM2_PWR]
SUSCLK[VCC_OPT]
GND(x11)
J601 M.2 Key−B WWAN
TP601 S3_M2_GPIO6
TP602 S3_M2_GPIO8
S3_M2_IFSEL
+1V8
R608 10k
R609 10k
+3V7_S3
C610 1uF
C611 100nF
C612 22pF
C613 22pF
C614 22pF
NC D1+ D1− NC
NC D2+ D2− NC
GND
U602 TPD4E05U06DQAR
VCC(C1)
RST(C2)
CLK(C3)
I/O(C7)
CD(NO_to_GND)
VPP(C6)
GND(C5)
SHIELD
J602 NSIM−2−C
H601 3052 standoff
H602 3042 standoff

SLOT 3: generic M.2 3052/3042 Key−B WWAN socket. Pins checked vs Sierra EM92XX PTS Rev 5 (Jan 2025) + Telit FN990 HDG Rev 3.
PCIe Gen3 x1 to switch port 5 (lane3); USB2 + USB3 to USB5744 port 3. Module TX AC caps on module.
FULL_CARD_POWER_OFF# (pin 6): TCA6416A 3.3V push−pull + 47k PD (EM92 VIH 1.35−4.4V; FN990 1.8V/VPH domain, 47k internal PD) −> module off by default.
  Firmware: drive FCPO# LOW before removing +3V7 (no high levels into an unpowered card − FN990 HDG 6.2.1 back−power note).
RESET# (pin 67) and W_DISABLE1# (pin 8): open−drain 2N7002 only (both vendors forbid pull−up/push−pull on RESET#). Pulse >=1s covers EM92 (450ms) + FN990 (1s).
PERST# = PCIE_PERST_N AND S_PCIE_EN_L (SN74LVC1G08, 3.3V; EM92 VIH 1.5V..VCC, FN990 VPH domain). 10k PD (rev 0.3, was PU) keeps PERST# low until
  firmware sets PCIE_EN >=150ms after the slot rail is valid (EM92 T_pvpgl; no host signal into the card for the first 100ms). JP = hard PERST# hold.
Pin 20 host−IF select: NFET pulls pin 20 LOW while SLOTn_PCIE_SEL is HIGH (10k PU = default PCIe). Own expander line (U802 ex−spare), NOT PCIE_EN:
  pin 20 is sampled at boot, while PERST# must stay gated low until >=150ms after power valid (EM92 T_pvpgl) − one line cannot do both.
  FN990 USB_PCIE_SWITCH (int. 10k PU to 1.8V): LOW = PCIe (+USB debug), HIGH = USB only. EM92 PCIE_DIS (PD, max 2.1V): LOW/open = PCIe, HIGH = USB
  −> fit the DNP 10k to +1V8 only to let PCIE_SEL=0 select USB on a Sierra card. Changing mode needs a module power cycle.
Pin 22: EM92 VBUS_SENSE (PD, 1.6−5.25V = USB present) / FN990 TGPIO_06 (1.8V). DNP 10k to slot rail: FIT ONLY for Sierra; NEVER with Telit.
Pin 28: EM92 PLA_S2_N (power−loss ack, LOW = safe to remove power) / FN990 TGPIO_08 −> test point (1.8V; not level−shifted in rev 0.3).
Pins 24/38/68 NC (EM92 optional VCC; FN990 TGPIO). Pin 73 = GND (both vendors) −> 3.3V sideband. 40−48 (UIM2), 56/58, 59−65, 23, 25, 26, CONFIG_x: NC.
SIM: CUI NSIM−2−C card−detect closes to GND when card present. FN990 UIM1_PRESENT: active−low OK. EM92 SIM1_DETECT default 0V=absent: set polarity by AT
  (configurable per PTS Table 2−1 note e) or disable hot−swap detect. ESD + 22pF filters near holder.
GNSS / NTN antennas: NO carrier RF or GNSS bias circuit. All antennas are off−board pigtails on the card U.FL/MHF4 ports (see docs/antenna−gnss−plan.md).
  FN990: internal GNSS bias LDO (2.9−3.15V, 100mA) on its GNSS port − active antenna OK. EM9291: GNSS on shared ANT1 − DC on ANT0−3 damages it: PASSIVE only.
Mechanical: standoff holes for 3052 (52mm) and 3042 (42mm) cards.

Datasheet−verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto−routed); rails = power symbols; inter−sheet = hier. labels
Jorge Mendoza − WWAN carrier (spin 1 bench prototype)
Sheet: /M.2 Key−B WWAN slot 3 (EM9291/FN990) + nano−SIM/
File: slot3.kicad_sch
Title: M.2 Key−B WWAN slot 3 (EM9291/FN990) + nano−SIM
Size: A2
Date: 2026−10−10
Rev: 0.3
KiCad E.D.A. 9.0.2
Id: 7/9


J701
USB-C (USB3.2 Gen1, UFP)
VBUS
CC1
CC2
D-
D+
RX1-
RX1+
TX1-
TX1+
RX2-
RX2+
TX2-
TX2+
SBU1
SBU2
SHIELD
GND
PWR_FLAG
D701
SMAJ5.0A
C702
1uF/16V
FB701
BLM18PG221SN1D
C703
1uF/16V
C704
100nF
+3V3
C705
100nF
U704
HD3SS3220RNHR
VDD5
VCC33
CC1
CC2
TXp
TXn
RXp
RXn
DIR
ID
INT_N/OUT3
VCONN_FAULT_N
SDA/OUT1
SCL/OUT2
RX1n
RX1p
TX1n
TX1p
RX2n
RX2p
TX2n
TX2p
VBUS_DET
ADDR
CURRENT_MODE
GND
PORT
ENn_CC
ENn_MUX
TP701
HD_DIR
R703
200k
R702
910k
R701
1M
C701
4.7nF/100V
U701
TPD4EUSB30DQAR
U702
TPD4EUSB30DQAR
U703
TPD4EUSB30DQAR
NC
D1+
D1-
D2+
D2-
C706 100nF
C707 100nF
C714
18pF
C715
18pF
R705
1M
Y701
25MHz
R704
12k 1%
TP702
HUB_RESET_N
HUB_RESET_N
R706
10k
C716
100nF
R707
49.9k
R708
100k
R709
10k
R710
10k
R711
200k
R712
10k
I2C_SCL
I2C_SDA
U705
USB5744/2G9 (verify ordering suffix)
VDD33
VDD12
USB2UP_DM
USB2UP_DP
USB3UP_TXDP
USB3UP_TXDM
USB3UP_RXDP
USB3UP_RXDM
XTALI/CLK_IN
XTALO
RBIAS
ATEST
RESET_N
VBUS_DET
SPI_CLK/SMCLK
SPI_DO/SMDAT
SPI_DI/CFG_BC_EN
SPI_CE_N/CFG_NON_REM
PRT_CTL1
PRT_CTL2
PRT_CTL3
PRT_CTL4/GANG_PWR
VSS(EP)
USB2DN_DP1/PRT_DIS_P1
USB2DN_DM1/PRT_DIS_M1
USB3DN_TXDP1
USB3DN_TXDM1
USB3DN_RXDP1
USB3DN_RXDM1
USB2DN_DP2/PRT_DIS_P2
USB2DN_DM2/PRT_DIS_M2
USB3DN_TXDP2
USB3DN_TXDM2
USB3DN_RXDP2
USB3DN_RXDM2
USB2DN_DP3/PRT_DIS_P3
USB2DN_DM3/PRT_DIS_M3
USB3DN_TXDP3
USB3DN_TXDM3
USB3DN_RXDP3
USB3DN_RXDM3
USB2DN_DP4/PRT_DIS_P4
USB2DN_DM4/PRT_DIS_M4
USB3DN_TXDP4
USB3DN_TXDM4
USB3DN_RXDP4
USB3DN_RXDM4
C708 100nF
C709 100nF
C710 100nF
C711 100nF
C712 100nF
C713 100nF
SLOT1_USB2_DP
SLOT1_USB2_DM
SLOT1_USB3_RX_P
SLOT1_USB3_RX_N
SLOT1_USB3_TX_P
SLOT1_USB3_TX_N
SLOT2_USB2_DP
SLOT2_USB2_DM
SLOT2_USB3_RX_P
SLOT2_USB3_RX_N
SLOT2_USB3_TX_P
SLOT2_USB3_TX_N
SLOT3_USB2_DP
SLOT3_USB2_DM
SLOT3_USB3_RX_P
SLOT3_USB3_RX_N
SLOT3_USB3_TX_P
SLOT3_USB3_TX_N
R713 10k
R714 10k
+3V3
C717
100nF
C718
100nF
C719
100nF
C720
100nF
C721
10uF/6.3V
+1V2
C722
100nF
C723
100nF
C724
100nF
C725
100nF
C726
100nF
C727
100nF
C728
100nF
C729
100nF
C730
10uF/6.3V
C731
10uF/6.3V

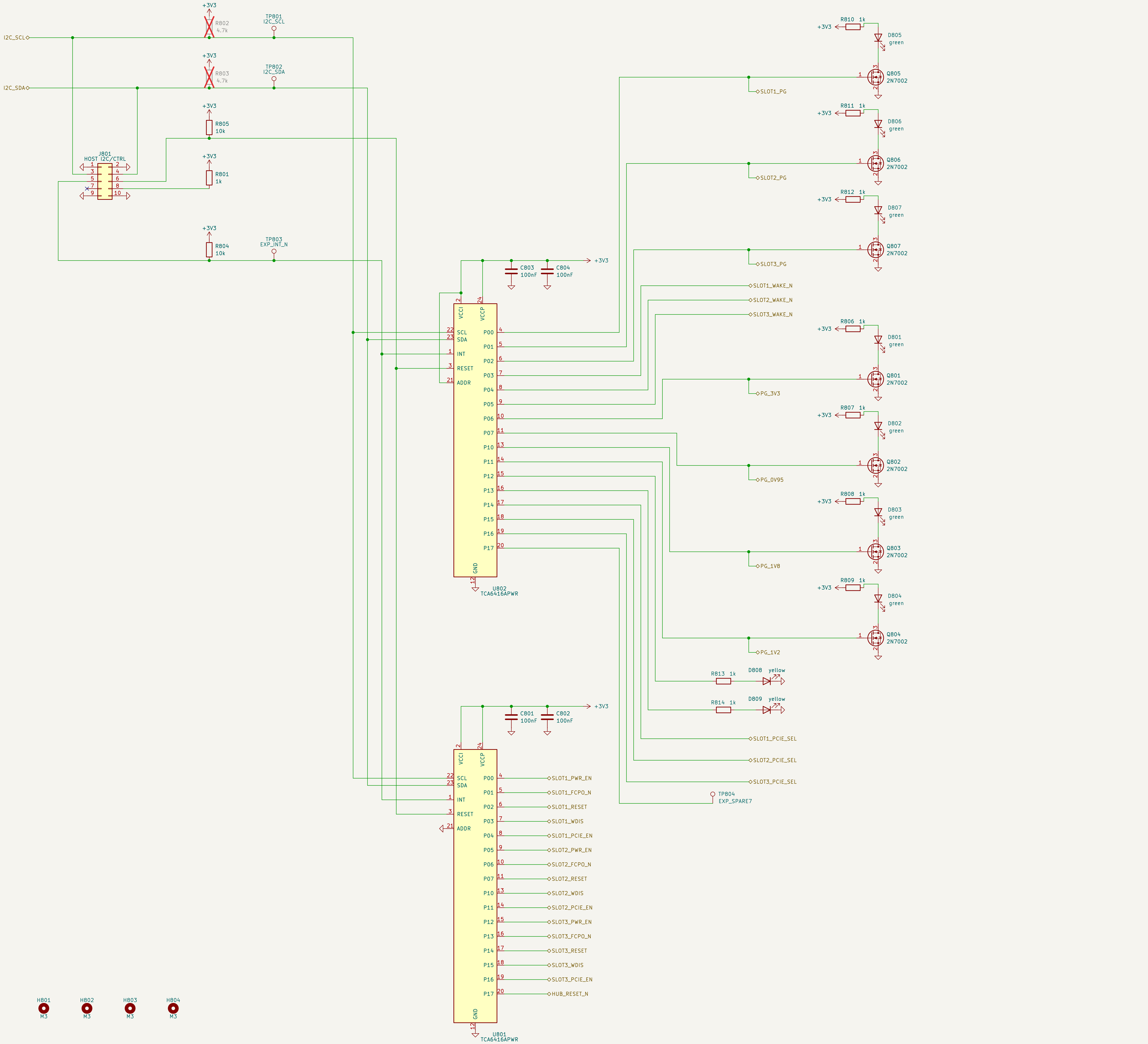
USB5744 (Microchip DS00001855J): SMBus slave mode (10k PU on SMCLK/SMDAT, sec 3.4.1), CFG_NON_REM 10k PU = ports 1–3 non–removable,
CFG_BC_EN 200k PD = no battery charging, port 4 disabled (PRT_DIS_P4/M4 to 3.3V). VBUS_DET via 49.9k/100k divider (detachable upstream).
VDD12 from +1V2 buck (688mA typ, 4 SS ports). RBIAS 12k 1%. 25MHz crystal. PRT_CTLx unused (slot power is switched by the 3.7V bucks).
Upstream USB–C (UFP only): HD3SS3220 PORT=GND (UFP), ADDR=NC (GPIO mode), ENn_CC/ENn_MUX=GND; it presents Rd and muxes SS by plug orientation.
VDD5 is taken from VBUS (UFP–only design; VERIFY with TI that VBUS–powered VDD5 is acceptable – else add a 5V rail).
Hub upstream TX AC caps 100nF between hub and mux; hub downstream TX caps 100nF on this sheet (slot RX).
ESD: TPD4EUSB30 on SS pairs + D+/D–/CC at the connector; SMAJ5.0A on VBUS.

Datasheet–verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto–routed); rails = power symbols; inter–sheet = hier. labels
Jorge Mendoza – WWAN carrier (spin 1 bench prototype)
Sheet: /USB3 hub USB5744 + USB–C upstream (HD3SS3220)/
File: usb_hub.kicad_sch

Title: USB3 hub USB5744 + USB–C upstream (HD3SS3220)

Size: A2	Date: 2026–10–10	**Rev: 0.3**
KiCad E.D.A. 9.0.2		Id: 8/9

Host header J801 (2x5 2.54mm) to FRDM-IMX95 3.3V I2C (VERIFY host header voltage is 3.3V; bus also hosts USB5744 SMBus and PCA9306->PI7C).
U801 TCA6416A @0x20 (ADDR=GND): outputs – per-slot PWR_EN, FCPO#, RESET (active-high to NFET), WDIS (active-high to NFET), PCIE_EN; HUB_RESET_N.
U802 TCA6416A @0x21 (ADDR=VCCP): inputs – slot PG, slot WAKE#, rail PGs; outputs – 2 user LEDs, SLOT1-3_PCIE_SEL (M.2 pin 20, rev 0.3). P17 spare (test pad).
All TCA6416A pins are inputs after POR –> board pull-ups/pull-downs define safe defaults (slots OFF, modules held in FCPO#).
Other I2C addresses: USB5744 SMBus (default 0x2D, VERIFY), PI7C switch 0x6F (via PCA9306). HD3SS3220 I2C disabled (GPIO mode).
Bus pull-ups: 10k on hub sheet (strap); 4.7k here DNP (host board normally provides).
PG LEDs: 2N7002 driven from each open-drain PG net (LED on = rail good).


I2C_SCL
I2C_SDA
+3V3
R802
4.7k
R803
4.7k
TP801
I2C_SCL
TP802
I2C_SDA
R805
10k
R801
1k
J801
HOST I2C/CTRL
R804
10k
TP803
EXP_INT_N
C803
100nF
C804
100nF
U802
TCA6416APWR
VCCI
VCCP
SCL
SDA
INT
RESET
ADDR
GND
P00
P01
P02
P03
P04
P05
P06
P07
P10
P11
P12
P13
P14
P15
P16
P17
SLOT1_PG
SLOT2_PG
SLOT3_PG
SLOT1_WAKE_N
SLOT2_WAKE_N
SLOT3_WAKE_N
PG_3V3
PG_0V95
PG_1V8
PG_1V2
R810 1k
D805
green
Q805
2N7002
R811 1k
D806
green
Q806
2N7002
R812 1k
D807
green
Q807
2N7002
R806 1k
D801
green
Q801
2N7002
R807 1k
D802
green
Q802
2N7002
R808 1k
D803
green
Q803
2N7002
R809 1k
D804
green
Q804
2N7002
R813 1k
D808 yellow
R814 1k
D809 yellow
SLOT1_PCIE_SEL
SLOT2_PCIE_SEL
SLOT3_PCIE_SEL
TP804
EXP_SPARE7
C801
100nF
C802
100nF
U801
TCA6416APWR
SLOT1_PWR_EN
SLOT1_FCPO_N
SLOT1_RESET
SLOT1_WDIS
SLOT1_PCIE_EN
SLOT2_PWR_EN
SLOT2_FCPO_N
SLOT2_RESET
SLOT2_WDIS
SLOT2_PCIE_EN
SLOT3_PWR_EN
SLOT3_FCPO_N
SLOT3_RESET
SLOT3_WDIS
SLOT3_PCIE_EN
HUB_RESET_N
H801
M3
H802
M3
H803
M3
H804
M3

Datasheet–verified pinouts noted per sheet; items marked VERIFY are not yet confirmed
Generated schematic (auto–routed); rails = power symbols; inter–sheet = hier. labels
Jorge Mendoza – WWAN carrier (spin 1 bench prototype)
Sheet: /Control: I2C GPIO expanders, host header, PG and status LEDs/
File: control.kicad_sch

Title: Control: I2C GPIO expanders, host header, PG and status LEDs

Size: A2	Date: 2026–10–10	**Rev: 0.3**
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